

The code for `$a_2$` is:

Please provide a brief explanation of your answer.

Evaluating this limit gives $a_2 = 1 + \varepsilon$, where ε is an astronomically small positive

number (involving terms like $72^{-10^{24}}$). Since $a_2 > 1$ strictly, the problem of minimizing total time $x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$ and $x_1, x_2 \geq 0$ has the unique optimal solution $(x_1^*, x_2^*) = (0, 1/a_2)$, with $1/a_2$ being incredibly close to (but strictly less than) 1.

The infinity-norm distance from $(0.3, 0.7)$ to the minimizer is $\max(0.3, |0.7 - 1/a_2|) \approx 0.3$, which is indeed at most 0.4.

The infinity-norm distance from $(0.6, 0.4)$ to the minimizer is $\max(0.6, |0.4 - 1/a_2|) \approx 0.6$, which is **not** at most 0.45.